

SimpleEnvelope.h:

AttRelEnvelope

- Set att, rel, samplerate
- Run:
 - 1: Look at prev input for att or rel
 - 2: run EnvelopeDetector

SimpleEnvelope.h:

EnvelopeDetector

- Set Tc, samplerate
- SetCoef:
 - $\text{coef_} = \exp(-1000.0 / (\text{ms_} * \text{samplerate_}))$;
- Run:
 - $\text{state} = \text{in} + \text{coef_} * (\text{state} - \text{in})$

SimpleComp.h:

SimpleComp

- Set thresh, ratio (in dB)
- initRunTime:
 - $\text{envdB_} = \text{DC_OFFSET}$;
- Process (stereo):
 - 1: rectify just in case
 - 2: add DC offset to avoid $\log(0)$
 - 3: lin2dB
 - 4: calc delta over threshold (> 0)
 - 5: add DC offset again
 - 6: run AttRelEnvelope
 - 7: subtract DC offset
 - 8: $\text{gr} = \text{envOutdB} * (\text{ratio}-1)$ (transfer func)
 - 9: dB2Lin
 - 10: apply to input

SimpleGain.h

- Lin2db
- dB2Lin

Credits:

2006, ChunkWare Music Software,
OPEN-SOURCE

SimpleComp.h:

SimpleCompRMS

- Set averagerMS, samplerate
- initRunTime:
 - $\text{envdB_} = \text{DC_OFFSET}$;
 - $\text{aveOfSqrS_} = \text{DC_OFFSET}$;
- Process (stereo):
 - 1: square inputs and sum them (+ DC offset)
 - 2: Run average's Envelope
 - 3: $\text{rms} = \sqrt{\text{answer step 2}}$
 - 4: process SimpleComp with inputs and rms

There are 3 envelopedetectors in total:

1. The averager
2. The attack
3. The decay

First average the signal

Then compare the signal to the previous state and decide att or dec